

HANDS-ON ACTIVITY WALKTHROUGH

Human TeleOp Flow

Students physically become the Java program and pass a token through INIT, PLAY, the active loop, and STOP.

20 MIN ACTIVITY	8 STUDENT ROLES	0 LIVE POWER NEEDED	1 PROGRAM TOKEN
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ACTIVITY OUTCOME Students explain what happens during INIT, identify where the program waits, trace what repeats after PLAY, and show how STOP ends the loop.

Materials

- Lesson 1 cutout cards: IDENTIFY, CONNECT, WAIT, READ, SET POWER, TELEMETRY, REPEAT, and STOP.
- One program token: a large plastic bolt, ball, marker, or other clearly visible object.
- One gamepad; TRUE and FALSE cards; several left/right joystick-value cards.
- Optional: motor pictures or paper labels for leftMotor and rightMotor.

The complete route

IDENTIFY Name the OpMode	CONNECT Map the motors	WAIT Hold for PLAY	READ Get joystick values
SET POWER Command motors	TELEMETRY Report values	REPEAT Return to READ	STOP Exit and disable

The token moves left to right through startup. After PLAY, REPEAT returns the token to READ. STOP sits outside the loop as the exit.

Student setup and roles

Place the students in a wide U shape. Put STOP slightly outside the READ–SET POWER–TELEMETRY–REPEAT loop. The student holding the program token is the only student allowed to perform a program action.

STUDENT	CARD	JOB	SAMPLE LINE
1	IDENTIFY	Announce the selected OpMode	"Basic Two Motor TeleOp begins at runOpMode."
2	CONNECT	Match Java names to configured hardware	"leftMotor maps to left_motor; rightMotor maps to right_motor."
3	WAIT	Hold execution until PLAY	"The program is waiting at waitForStart."
4	READ	Read the newest joystick positions	"The newest left and right values are..."
5	SET POWER	Send calculated values to the motors	"I command leftMotor and rightMotor."
6	TELEMETRY	Report the commanded values	"The Driver Station shows..."
7	REPEAT	Check whether the OpMode is active	"True repeats; false exits."
8	STOP	End the loop and confirm disabled	"The OpMode ended; outputs are disabled."

Ground rules to tell students

- Perform your action only while holding the program token.
- Say your line loudly enough for the group to hear.
- Do not skip a station or pass the token backward unless REPEAT sends it to READ.
- When STOP receives the token, all activity ends immediately.

Twenty-minute teacher walkthrough

1. Introduce the simulation | 2 minutes

TEACHER	<i>"You are about to become the Java program. This token shows which instruction the Robot Controller is currently executing. You may act only while holding it."</i>
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STUDENT	<i>"Students take their assigned positions. IDENTIFY receives the program token."</i>
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ASK	<i>"Does a Java program perform every instruction at the same time?"</i>
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LISTEN FOR	No. It follows instructions in an order.
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TEACHING NOTE	Keep the token large and visible. If students act out of turn, pause and ask who currently owns execution.
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2. Walk through INIT | 4 minutes

TEACHER	<i>"The driver selects Basic Two Motor TeleOp and presses INIT. The program begins."</i>
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STUDENT	<i>"IDENTIFY: 'This is Basic Two Motor TeleOp. The program begins at runOpMode.' Pass the token to CONNECT. CONNECT: 'I use hardwareMap to connect leftMotor to left_motor and rightMotor to right_motor.' Pass to WAIT. WAIT: 'Setup is complete. The program is waiting at waitForStart.' Keep the token."</i>
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ASK	<i>"Can READ take the token yet?"</i>
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LISTEN FOR	No. PLAY has not been pressed, so execution remains at WAIT.
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TEACHING NOTE	Let READ move the gamepad before PLAY. Nothing should advance. This makes the waiting state memorable.
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3. Rehearse POWERING ON | 3 minutes

TEACHER	<i>"Robot is secured and disabled. I am looking around to make sure everyone is clear. POWERING ON!"</i>
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STUDENT	<i>"WAIT acts as the Driver Station operator. The other students visibly step behind the safety boundary. WAIT pauses, checks the area, says "The area is clear. Pressing PLAY," and passes the token to READ."</i>
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ASK	<i>"What must happen between the callout and pressing PLAY?"</i>
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LISTEN FOR Everyone gets time to move; the operator visually confirms the area is clear.

TEACHING NOTE If the student rushes or skips the pause, stop the activity and restart the entire call-and-clear sequence.

NON-NEGOTIABLE SEQUENCE Secure and disable → look around → call "POWERING ON!" → give time to step away → confirm clear → press PLAY.

4. Run the first loop cycle | 4 minutes

TEACHER	<i>"Both sticks are pushed forward. Trace one complete cycle and predict what the robot would do."</i>
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STUDENT	<i>"READ receives raw left Y = -0.6 and raw right Y = -0.6, then says the program negates them to +0.6 and +0.6. SET POWER commands +0.6 to both motors. TELEMETRY reports Left Power: 0.6 and Right Power: 0.6. REPEAT asks whether the OpMode is active."</i>
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ASK	<i>"What movement do we predict?"</i>
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LISTEN FOR With correctly configured directions, equal positive values should move the robot straight forward. Then hold up TRUE: REPEAT returns the token directly to READ, not to IDENTIFY or CONNECT.

5. Change the joystick commands | 3 minutes

TEACHER	<i>"The driver changed the sticks. The loop must use the newest values."</i>
STUDENT	<i>"Cycle two: READ receives +0.7 left power and +0.3 right power. The group traces READ → SET POWER → TELEMETRY → REPEAT and predicts a curve toward the slower right side. Cycle three: both powers are 0.0; the group traces the cycle again."</i>
ASK	<i>"Does 0.0 motor power mean the OpMode stopped?"</i>
LISTEN FOR No. The motors are commanded to stop, but the active loop continues until STOP is requested.	
TEACHING NOTE This is the key distinction between a stopped mechanism and a stopped OpMode.	

6. Press STOP during the loop | 2 minutes

TEACHER	<i>"The driver pressed STOP. Check the OpMode state before another cycle."</i>
STUDENT	<i>"At REPEAT, the class answers FALSE. REPEAT passes the token to STOP instead of READ. STOP says: 'The loop exits. Motor outputs are disabled. The OpMode has ended.' STOP keeps the token."</i>
ASK	<i>"Can READ take another joystick value now?"</i>
LISTEN FOR No. The loop ended and no new iteration begins.	
TEACHING NOTE Nobody moves or speaks after STOP takes the token. The stillness reinforces that execution has ended.	

Debrief

Point to cards as students answer. Ask them to use the card names in their explanations.

ASK	LISTEN FOR
Which stages happen only during startup?	IDENTIFY, CONNECT, and WAIT.
Where does the program wait for PLAY?	At waitForStart(), represented by WAIT.
Which stages repeat?	READ, SET POWER, TELEMETRY, and REPEAT/active check.
What sends information to the Driver Station?	TELEMETRY.
What happens when the active check becomes false?	The token leaves the loop and goes to STOP.
Is zero motor power the same as pressing STOP?	No. Zero requests no motor output; STOP ends the OpMode.
CLASS CLOSING LINE Identify, connect, wait, read, set power, telemetry, repeat—and STOP exits.	

Optional error round

Run the startup again, but give CONNECT a misspelled configuration name such as left motor instead of left_motor.

CONNECT	<i>"Hardware mapping failed. The configured device could not be found."</i>
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TEACHER	<i>"Why can't we press PLAY and drive?"</i>
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LISTEN FOR	Initialization did not finish, so the program never reached waitForStart() or the control loop.
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CONNECT keeps the token. Do not allow it to advance to WAIT. This makes an initialization failure visible and concrete.

Optional real-robot extension

SAFETY CONDITION Use a stable robot stand with both wheels clear. Confirm DISABLED before approach and repeat the full POWERING ON call-and-clear sequence before PLAY.

- Secure the robot on blocks and confirm both wheels are clear.
- Select the OpMode and press INIT. Ask which student stages just occurred.
- Identify the current WAIT state at waitForStart().
- Perform the complete POWERING ON sequence and press PLAY only when clear.
- Move one stick gently and compare the wheel response with the human simulation.
- Watch live telemetry and ask which student role it represents.
- Press STOP and confirm DISABLED before anyone approaches the robot.

Completion check

INDIVIDUAL ASSESSMENT Hand the program token to any student and ask: “Where are you in the program, what happens next, and does this step run once or repeat?”

Evidence of success

- The student correctly names the current program phase.
- The student identifies the next destination for the token.
- The student distinguishes startup actions from repeated loop actions.
- The student explains that FALSE or STOP exits the loop.
- The student performs the POWERING ON sequence in the correct order.

Common misconceptions to correct

- Declaring a motor is not the same as mapping it to hardware.
- INIT does not start the driving loop.
- Centered joysticks stop motor commands but do not end the OpMode.
- Telemetry reports commanded values; it does not prove actual wheel speed or direction.
- STOP ends execution; troubleshooting begins only after DISABLED is confirmed.